

A Common Gustatory and Interoceptive Representation in the Human Mid-Insula

Jason A. Avery,^{1,2} Kara L. Kerr,^{1,3} John E. Ingeholm,⁴ Kaiping Burrows,¹
Jerzy Bodurka,^{1,5,6} and W. Kyle Simmons^{1,7*}

¹Laureate Institute for Brain Research, Tulsa, Oklahoma 74136

²Department of Biological Sciences, The University of Tulsa, Tulsa, Oklahoma 74104

³Department of Psychology, The University of Tulsa, Tulsa, Oklahoma 74104

⁴Laboratory of Brain and Cognition, National Institute of Mental Health (NIMH), National Institutes of Health (NIH), Bethesda, Maryland 20892

⁵College of Engineering, The University of Oklahoma, Norman, Oklahoma 73071

⁶Center for Biomedical Engineering, The University of Oklahoma, Norman, Oklahoma 73071

⁷Department of Community Medicine, The University of Tulsa, Tulsa, Oklahoma 74104

Abstract: The insula serves as the primary gustatory and viscerosensory region in the mammalian cortex. It receives visceral and gustatory afferent projections through dedicated brainstem and thalamic nuclei, which suggests a potential role as a site for homeostatic integration. For example, while human neuroimaging studies of gustation have implicated the dorsal mid-insular cortex as one of the primary gustatory regions in the insula, other recent studies have implicated this same region of the insula in interoception. This apparent convergence of gustatory and interoceptive information could reflect a common neural representation in the insula shared by both interoception and gustation. This idea finds support in translational studies in rodents, and may constitute a medium for integrating homeostatic information with feeding behavior. To assess this possibility, healthy volunteers were asked to undergo fMRI while performing tasks involving interoceptive attention to visceral sensations as well as a gustatory mapping task. Analysis of the unsmoothed, high-resolution fMRI data confirmed shared representations of gustatory and visceral interoception within the dorsal mid-insula. Group conjunction analysis revealed overlapping patterns of activation for both tasks in the dorsal mid-insula, and region-of-interest analyses confirmed that the dorsal mid-insula regions responsive for visceral interoception also exhibit strong responses to tastants. *Hum Brain Mapp* 00:000–000, 2015. © 2015 Wiley Periodicals, Inc.

Key words: gustation; interoception; insula

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*Correspondence to: Kyle Simmons, PhD, Laureate Institute for Brain Research, 6655 South Yale Ave., Tulsa, Oklahoma 74136 - 3326. E-mail: wksimmons@laureateinstitute.org

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INTRODUCTION

Situated within the Sylvian Fissure and hidden beneath the folds of the frontoparietal and temporal operculum, the insular cortex serves as primary viscerosensory cortex, receiving visceral afferent projections from the vagus nerve via dedicated thalamocortical relays [Craig, 2002]. Via a parallel neural pathway, the insula also receives gustatory afferents through cranial nerves VII, IX, and X [Beckstead et al., 1980; Pritchard et al., 1986]. Functional neuroimaging data accord well with this regions' anatomical connectivity, as numerous neuroimaging studies and meta-analyses have reported that the human insula supports both viscerosensory and gustatory representations [Pollatos et al., 2007; Small, 2010; Veldhuizen et al., 2011; Wang et al., 2008]. Because of its anatomical connections and functional properties, it has been suggested that the insular cortex is responsible for integrating visceral interoceptive and gustatory information in the service of higher-order homeostatic regulation [De Araujo et al., 2012]. A common gustatory and interoceptive representation within the human insular cortex could serve as a means by which metabolic and ingestive factors modulate feeding behavior in service of homeostasis. An example of this may come from the recent demonstration that the response of gustatory cortex in the dorsal mid-insula to food pictures is negatively correlated with levels of circulating blood glucose [Simmons et al., 2013b]. This dynamic modulation of the response to food stimuli by signals of peripheral energy availability may result from the integration of gustatory and interoceptive signals within the dorsal mid-insula. This account of these findings would be strengthened by evidence that human gustatory cortex is indeed sensitive to interoceptive signals. Unfortunately, direct evidence to this effect in humans has not heretofore existed.

Translational evidence from rodent electrophysiology studies supports the notion of overlapping projection fields for gustatory and visceral stimulation within the rodent insular cortex, with many neurons exhibiting excitatory responses to both interoceptive and gustatory stimulation [Hanamori et al., 1998]. Neurons within rodent gustatory cortex also appear to respond to the post-ingestive effects of sucrose, supporting the functional role of this region in both feeding and homeostasis [Oliveira-Maia et al., 2012]. Although there is currently no primate data to directly support gustatory-interoceptive overlap, there is limited indirect evidence, with separate human neuroimaging studies employing either gustatory or interoceptive stimuli observing activity in the same vicinity of the dorsal mid-insula [Pollatos et al., 2007; Simmons et al., 2013a, 2013b; Small, 2010; Veldhuizen et al., 2011; Wang et al., 2008]. Unfortunately, the overlap in these separate studies does not suffice as evidence of direct overlap, as differences in image acquisition, spatial smoothing, as well as insular functional organization between different groups of subjects may cause activations within neighboring but distinct regions of dorsal mid-insula to appear to reside in the same cortical location. At present, no study has

directly compared the cortical activity for gustation and interoception within the same group of individuals.

Healthy volunteers were thus recruited and asked to undergo fMRI while performing tasks designed to map gustatory and visceral interoceptive processing in the insula. Based on prior neuroimaging and translational evidence, it was hypothesized that the mid-insula would exhibit overlapping activation patterns for gustation and interoception.

METHODS

Participants

Twenty right-handed, native English-speaking volunteers (8 Female; Age: Mean (SD) = 28 (7), Range = 18–39; Body-Mass-Index (BMI): Mean (SD) = 29 (6), Range = 20–43) participated in the study. All subjects underwent clinical assessment prior to participating in the study including the Structured Clinical Interview for DSM-IV Axis-I Disorders (SCID-I) conducted by trained Master's or Doctorate level clinicians with experience in psychiatric diagnosis. Volunteers were excluded from participation for having met criteria for any Axis-I psychiatric disorder at any point in their lifetime. Subjects were also excluded if they had a history of neurological disorder, traumatic brain injury with loss of consciousness, psychotropic medication use, substance dependence, or current pregnancy. Additionally, subjects were excluded if they had taken any drugs likely to affect cerebral blood flow within the three weeks prior to scanning. All subjects were paid for their participation and provided written informed consent as approved by the University of Oklahoma Institutional Review Board.

Experimental Design

Each subject received a structural MRI scan followed by a series of functional MRI scans, during which they performed first the Gustatory Mapping (GM) task, followed by the Interoceptive Attention (IA) task (Fig. 1). Each subject performed both tasks on the same day, within the same scan session. Visual stimuli were projected onto a screen located inside the scanner bore and viewed through a mirror system mounted on the head-coil.

GM Task (Fig. 1a): This task involved three types of trials: *Cue* trials, *Tastant* trials, and *Wash/Swallow* trials. During *Cue* trials, the word "SWEET" or "NEUTRAL" appeared on the screen for five seconds. During *Tastant* trials, the word "TASTE" appeared on the screen for 5 s and then either the sweet tastant (0.4 mL of 0.6 M sucrose) or the neutral tastant (0.4 mL of distilled water) was delivered onto the subject's tongue. During *Wash/Swallow* trials, the word "WASH" appeared for 2.5 s and 0.8 mL of distilled water was delivered onto the subject's tongue. Immediately after this, the word "SWALLOW" appeared

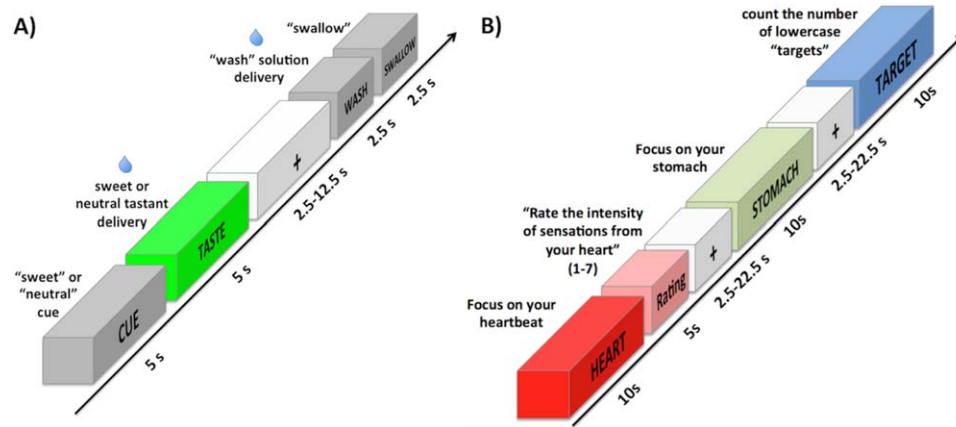


Figure 1.

fMRI task design. **(A)** Gustatory Mapping task. The delivery of a sweet or neutral tastant was preceded by a word cue and followed by a “wash/swallow” period to rinse out and remove the liquid. **(B)** Interoceptive Attention task. During 10-second trials,

subjects would focus on a part of the body indicated by a cue word, either “HEART”, “STOMACH”, or “BLADDER,” or would count the number of times (between 1 and 7) that the word “TARGET” switched to the lowercase “target.”

for 2.5 s, directing the subject to swallow. The verbal cues for each type of GM task trial were presented in black font against a white background. (For a detailed description of the GM task design, see the Supporting Information section.)

Outside of the scanner, following the completion of both GM and IA scan sessions, participants were asked to make ratings of the perceived sweetness, intensity, and pleasantness of the sweet and neutral tastants delivered during the Gustatory Mapping task. These items were rated on a scale ranging from 1 (not sweet/intense/pleasant) to 10 (extremely sweet/intense/pleasant). Tastant solutions were delivered via a pneumatically driven MR-compatible tastant delivery system. Solutions were kept at room temperature (approximately 20°C) in pressurized syringes and delivered to the participant via plastic tubing to a gustatory manifold attached to the head-coil. This manifold delivered solutions directly onto each subject’s tongue during scanning. LabView (National Instruments, Austin, TX) software controlled the precise timing and quantity of tastants dispensed to the subject during the scan. Each of the four GM task scans lasted 620 s.

IA Task (Fig. 1b): During this task, each subject completed three 550 s fMRI scans throughout which they performed two experimental conditions: the interoception condition and the exteroception control condition. During the interoception condition, the word “HEART”, “STOMACH”, or “BLADDER” was presented for 10 s in black font against a white background, during which time subjects were instructed to focus attention on the intensity of the sensations experienced from that organ. For example, upon seeing the word “HEART,” subjects focused on how intensely they could feel the sensation of their heart beating. By requiring participants to focus their attention

on internal sensations from their heart and viscera, this interoception condition makes use of the attentional spotlight effect, demonstrated in other sensory modalities such as touch and taste [Johansen-Berg et al., 2000; Veldhuizen et al., 2007], to amplify the signal within cortical regions underlying viscerosensory perception. It is now well established that this task is effective at mapping interoceptive insular cortex [Avery et al., 2014; Simmons et al., 2013a].

As an exteroceptive control condition, subjects fixated on the word “TARGET” which randomly switched to the lowercase “target” for 500 ms durations during the 10 s exteroceptive task trial. Subjects were instructed to attend to the exteroceptive target and to count the number of times they saw the lowercase word during each trial. After half of the trials of each condition, subjects were given 5 s response periods to rate the intensity of interoceptive sensations (with “1” indicating little sensation, “4” indicating moderately intense sensation, and “7” indicating an extremely intense sensation), or indicate the number of targets counted, during the previous trial, using an MR-compatible scroll-wheel. These rating periods were included to insure that participants remained attentive to the task. Each condition trial was separated by a variable-duration interstimulus interval lasting between 2.5 and 22.5 s (mean interval = 6.7 s), during which time subjects saw only a black fixation mark against a white background. After receiving verbal instructions, all subjects practiced the interoception and exteroception tasks prior to performing them in the scanner, were observed to make stimulus intensity responses, and finally were asked to indicate whether they had any remaining questions about the task demands.

In each of the fMRI task scanning runs, both the GM and IA task conditions were presented in a pseudo-random order optimized for fMRI analysis by Optseq2

(<http://surfer.nmr.mgh.harvard.edu/optseq/>). Stimulus presentation and response collection were controlled using Eprime2 software (www.pstnet.com).

Imaging

Functional and structural MR images were collected using a General Electric Discovery MR750 (GE Healthcare, Milwaukee, WI) whole-body 3-Tesla MRI scanner, using a scalable 32-channel digital MRI receiver capable of performing massively parallel fMRI. A brain-dedicated receive-only 32-element coil array (Nova Medical Inc, Wilmington, MA), optimized for parallel imaging, was used for MRI signal reception. A single-shot gradient-recalled echo-planar imaging (EPI) sequence with Sensitivity Encoding (SENSE) depicting blood oxygenation level-dependent (BOLD) contrast was used for functional scans. A T1-weighted magnetization-prepared rapid gradient-echo sequence with SENSE was used to provide an anatomical reference for the fMRI analysis.

EPI imaging parameters: FOV/slice/gap = 240/2.9/0 mm, slices/volume (axial) = 46, acquisition matrix = 96×96 , repetition/echo time TR/TE = 2500/30 ms, SENSE acceleration factor $R = 2$ in the phase encoding (anterior-posterior) direction, flip angle = 90° . EPI image matrix = 128×128 , fMRI voxel volume = $1.875 \times 1.875 \times 2.9 \text{ mm}^3$. Both IA and GM tasks used identical imaging parameters, with the exception of scan durations (IA task: 3 scans, 220 volumes/scan, 550 s/scan; GM task: 4 scans, 248 volumes/scan, 620 s/scan).

Anatomical Image: FOV = 240 mm, slices/volume (axial) = 176, slice thickness = 0.9 mm, image matrix = 256×256 , voxel volume = $0.938 \times 0.938 \times 0.9 \text{ mm}^3$, TR/TE = 5/2.02ms, acceleration factor $R = 2$, flip angle = 8° , inversion time = 725 ms, scan time = 372 s.

Image Preprocessing

Image preprocessing was performed using AFNI and SUMA (<http://afni.nimh.nih.gov/afni>). The anatomical scan was registered to the EPI time-course and then transformed into an anatomical surface model using FreeSurfer (<http://surfer.nmr.mgh.harvard.edu/>). Using the SUMA program MapIcosahedron, each subject's cortical surface, containing between 120,000 and 190,000 nodes/hemisphere, was transformed to a standardized cortical surface containing an identical number of nodes (156,252 nodes/hemisphere) and identical node indices across subjects [Argall et al., 2006]. The first 4 volumes of each EPI time-course (10 s) were excluded from data analysis to allow the fMRI signal to reach longitudinal equilibrium, and a slice timing correction was applied to all EPI volumes. All EPI volumes were registered to the base EPI volume (the first volume of the first EPI time-course) using a 6-parameter (3 translations, 3 rotations) motion correction algorithm, and the motion estimates were saved for use as regressors in the subsequent statistical analyses. Additionally, motion-censoring algorithms were

implemented to guard against potential artifacts induced by uncontrolled subject motion which may persist despite both volume registration and motion regression (i.e., motion scrubbing) [Power et al., 2012]. This procedure created a list of time points within each EPI time-series in which the euclidean-normalized derivative of the subject's motion parameters was greater than 0.3 (roughly 0.3 mm motion). This list was then provided to the AFNI program 3dDeconvolve, which censored those time points during the subsequent regression analysis. Any subject with a mean euclidean-normalized derivative of greater than 0.2 during *either* the GM or IA scans was excluded from the analysis. Following volume registration, the EPI data were then transformed and mapped to the standardized cortical surface, and the signal intensity for each EPI volume was normalized to reflect percent signal change from the mean intensity across the time-course.

Given the goals of this study, a comment on the use of surface analysis and preprocessing choices is warranted. First, functional data computed from standard volume-based analyses can become distorted by individual differences in cortical folding patterns [Van Essen and Drury, 1997]. The alignment of functional data across subjects is thus greatly enhanced by mapping to a standardized cortical surface model, which reduces intersubject variability in cortical folding patterns and preserves the topology of cortical surface areas among subjects [Argall et al., 2006]. Additionally, *no spatial smoothing was applied to the EPI data*, to further reduce the chances of artificially producing overlapping activation patterns via spatial averaging. As Gaussian smoothing is commonly used to increase the effective signal-to-noise ratio (SNR) within EPI datasets, this does lead to an appreciable loss of SNR in the present datasets, compared to more traditional preprocessing approaches. However, the more conservative approach employed within this study allows for a more precise and reliable localization of the cortical regions activated by gustatory and interoceptive tasks.

Statistical Analyses

Separate multiple linear regression models were constructed to examine the data from the two tasks. For the IA task, this model included regressors for each interoception condition and the exteroception condition, as well as regressors for the response periods following those conditions. To adjust the model for the shape and delay of the BOLD function, the task regressors were constructed by convolution of a gamma-variate function with a boxcar function having a 10 s width beginning at the onset of each trial period.

For the GM task the regression model included regressors for *Sweet Cue* and *Neutral Cue* trials, *Sweet Tastant* and *Neutral Tastant* trials, as well as *Wash/Swallow* trials. In order to separately model the response to the *Tastant* trials

from the response to the *Cue* trials and *Wash/Swallow* trials (see Fig. 1a and GM task description above) the regressors were constructed in the following manner: The *Sweet Cue* and *Neutral Cue* regressors included all of the free-standing *Cue* trials, as well as the *Cue* trials that occurred prior to the *Tastant* trials. The *Wash/Swallow* regressor included both the free-standing *Wash/Swallow* trials, as well as the *Wash/Swallow* trials that occurred 2.5–12.5 s after *Tastant* trials. The five task regressors were constructed by convolution of a gamma-variate function with a box-car function having a 5 s width beginning at the onset of each trial period.

The regression model for both tasks also included regressors of noninterest to account for each run's mean, linear, quadratic, and cubic signal trends, as well as the six normalized motion parameters (three translations, three rotations) computed during the volume registration preprocessing.

Additionally, paired-sample *t*-tests were performed on the tastant ratings participants completed outside of the scanner, to assess the participants' subjective experience of the tastants in the GM task.

Group-Level Analyses

Two approaches were taken to identify shared gustatory-interoceptive cortical representations. In the first, the individual subject regression coefficients and *t*-statistics created for each experimental condition were then combined in a group-level mixed-effects meta-analysis using the AFNI program 3dMEMA [Chen et al., 2012], which weighs individual subject regression coefficients by their reliability estimates (standard error). In this manner, 3dMEMA is able to take into account variability at both the subject and group level. For the IA task, the contrast *Interoception* vs. *Exteroception* was used to locate interoceptive regions of the insula. For the GM task, the contrast *Sweet Tastant* vs. *Neutral Tastant* was used to locate insula regions exhibiting significantly greater activity to the taste of sucrose than to distilled water. An initial *P*-value threshold of 0.005 was applied to both group contrast maps. The contrast maps for both tasks were separately corrected for multiple comparisons by cluster-size thresholding along the cortical surface, implemented via Monte Carlo simulations of cluster size by the SUMA program *slow_surf_clustsim.py*, to achieve correction for multiple comparisons at $P < 0.05$. Subsequently, a conjunction of the two corrected maps was created to locate cortical regions responsive to both gustation AND interoception.

In the second approach, a series of region-of-interest (ROI) analyses were performed to examine whether interoceptive regions of the brain also exhibit a significant gustatory response. As the insula's involvement in interoception and gustation was a primary motivation of this study, five ROIs within the insula that exhibited significant activation for either interoception or exteroception in the corrected IA task contrast (Figs. 2a and 3) were chosen for these ROI analyses. Within those ROIs, the average value of each subject's beta

coefficients for the sweet and neutral stimulus conditions from the GM task was extracted and a group-level pairwise *t*-test of those coefficients was performed. These ROI analyses were subsequently FDR corrected for multiple comparisons at $p_{\text{corrected}} < 0.05$ [Benjamini and Hochberg, 1995].

RESULTS

Behavioral Responses

Subjects' responses during the IA task indicated that they performed the tasks as instructed (See Table I). Responses to exteroceptive target detection trials were highly accurate [accuracy: Mean(SD)% = 87(15)%; Table I]. Ratings following interoceptive attention trials were slightly above the middle of the 7-point scale [Mean (SD): Heart = 4.1(1.3), Stomach = 5.1(1.6), Bladder = 5.5(1.6); Table I], indicating that participants experienced a moderately intense level of interoceptive sensation. Postscan ratings of the tastants delivered during the Gustatory Mapping task (See Table I) indicated that subjects perceived the taste of the sucrose as significantly sweeter ($t_{19} = 13.6$, $P < 0.001$) and more intense ($t_{19} = 7.82$; $P < 0.001$) than the neutral tastant. Pleasantness ratings for the sweet tastant were greater, but not significantly greater ($t_{19} = 1.5$, $P = 0.16$), than for the neutral tastant.

Imaging Results

Interoception

Within the insula, interoceptive attention resulted in significantly greater activation than exteroceptive attention bilaterally in mid and posterior regions of the insula (Fig. 2a; Table II). Notably, the mid-insula regions exhibiting greater activity for interoception were located within the fundus of the superior insular sulcus, in a region of the mid-insula previously identified as interoceptive-selective and homeostatically sensitive [Simmons et al., 2013a, 2013b]. The exteroception task resulted in significantly greater activation than interoception in right dorsal anterior insula, a finding that also replicates previously reported results [Simmons et al., 2013a]. Outside of the insula, significant activation clusters for interoception vs. exteroception were located in neighboring temporal cortex, medial frontal gyrus, and medial orbitofrontal cortex (Table II).

Gustation

Significant differences between *Sweet Tastant* vs. *Neutral Tastant* were observed in bilateral mid-insular cortex along the posterior short insular gyri and extending into the central insular sulcus (Fig. 2b; Table III), in agreement with meta-analysis of human neuroimaging studies of gustatory stimulation [Veldhuizen et al., 2011]. Outside of the insula, significant gustatory activations were observed within bilateral precentral and postcentral gyrus (Table III).

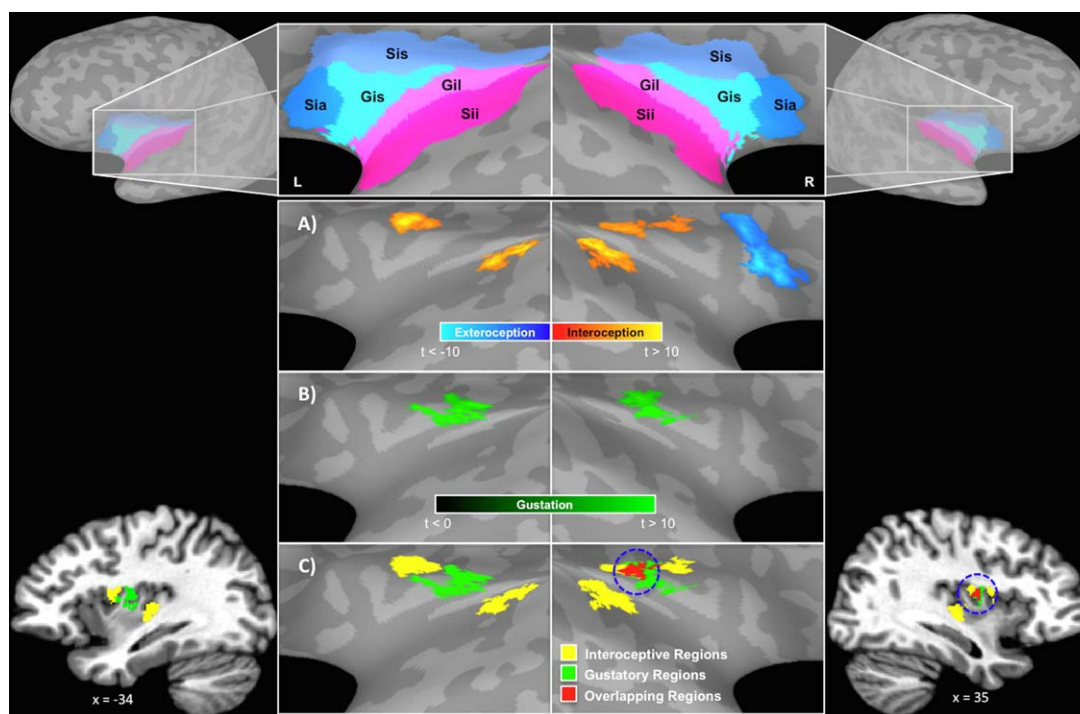


Figure 2.

Gustatory–Interceptive Overlap. The unsmoothed fMRI data was mapped and analyzed on a standardized cortical surface model (above), and statistical maps were cluster-size corrected for multiple comparisons at $P < 0.05$. **(A)** Within the insula, interoceptive attention (yellow-orange) resulted in significantly greater activation than exteroceptive attention (blue) bilaterally in mid and posterior regions of the insula. **(B)** Bilateral regions of the dorsal mid-insula also exhibited a reliably greater

response to the sweet vs. the neutral tastant. **(C)** A union of the statistical maps in A and B. Gustation and interoception tasks coactivated a region of right dorsal mid-insula, situated at the dorsal margin of the posterior short insular gyrus. (Bottom left and right) Volume renderings of the surface figures in C. Gil—long insular gyri, Gis—short insular gyri, L—left, R—right, Sia—anterior insular sulcus, Sii—inferior insular sulcus, Sis—superior insular sulcus.

Gustatory-Interceptive Overlap

The conjunction of gustatory and interoceptive contrast maps (Fig. 2c) revealed only one cortical region activated by both interoceptive and gustatory tasks, located in right dorsal mid-insula, near the intersection of the central insular sulcus with the superior insular sulcus. Subsequent transformation of this cluster back into volumetric space revealed that it lies in the same cortical location previously identified as selective for interoceptive awareness [Simmons et al., 2013a] (center-of-mass coordinates: [37, -5, 12] according to Talairach stereotaxic atlas [Talairach and Tournoux, 1988]) (Fig. 2c).

ROI Analyses

As described above, ROI analyses were conducted within the specific anterior, mid, and posterior insular clusters identified in the Interoception vs. Exteroception contrast (Fig. 3; Table IV). Within the dorsal anterior insula ROI (which showed a greater response to exteroception vs. interoception), there was no difference in activation

between sweet or neutral tastants ($t_{19} = 1.3$; $P = 0.22$), though both tastants elicited significant activation compared to baseline. The posterior insula interoception ROIs were unresponsive to both the sweet and neutral tastants ($P > 0.37$), and neither exhibited a difference between activation for sweet vs. neutral tastants ($P > 0.54$). In contrast, in both dorsal mid-insula interoceptive ROIs, sweet and neutral tastants produced significant activation above baseline ($P < 0.001$), and sweet tastants exhibited significantly greater activation than neutral tastants (left: $t_{19} = 3.2$, $P < 0.005$; right: $t_{19} = 4.3$, $P < 0.001$).

DISCUSSION

Prior meta-analytic evidence, combined with the results of recent neuroimaging studies in humans, suggests that the dorsal mid-insula is responsive to both gustatory and interoceptive stimuli, and that the responsiveness of this region to food stimuli is homeostatically sensitive [Kurth et al., 2010; Pollatos et al., 2007; Simmons et al., 2013a, 2013b; Small, 2010; Veldhuizen et al., 2011; Wang et al.,

TABLE I. Behavioral data

	Sweet	Neutral	t_{19}	P
Tastant ratings from gustatory mapping task^a				
Intensity	6.1(2.5)	2.3(1.8)	7.8	<0.001
Sweetness	7.5(1.5)	2.0(1.5)	13.6	<0.001
Pleasantness	6.1(2.9)	4.9(1.8)	1.5	0.16
Interoception ratings^b				
Condition	Mean (SD)			
Heart	4.1 (1.3)			
Stomach	5.1 (1.6)			
Bladder	5.5 (1.6)			
Exteroception % accuracy	86.9 (14.6)			

^aTastant ratings were collected after scanning, and were made on a 10-point scale ranging from 1 = not at all, 5 = somewhat, to 10 = extremely. Where appropriate, the table cells contain Mean(Standard Deviation) values.

^bInteroception ratings collected during the Visceral Interoceptive Awareness task were made on a 7-point scale ranging from 1 = little sensation, 4 = moderately intense sensation, to 7 = extremely intense sensation. Where appropriate, the table cells contain Mean(Standard Deviation) values.

2008]. Using a series of tasks previously demonstrated to identify insula regions responsive for gustation and visceral interoception [Simmons et al., 2013a, 2013b], group analyses were conducted to identify cortical regions responding to both functional tasks, using unsmoothed high-resolution fMRI data mapped to a standardized cortical surface, in the same individuals. The dorsal mid-insula was the only cortical region significantly responsive to both interoceptive attention as well as the taste of sucrose.

This is the first study to directly demonstrate the overlap between gustatory and interoceptive processing within the human brain. The present study's findings are in agreement with prior translational evidence of overlapping gustatory and visceral representation within rodent insular cortex [Hanamori et al., 1998]. Consistent with these findings in rodents, the region identified in this human study exhibiting gustatory and interoceptive responses resides in dorsal granular insular cortex [Evrard et al., 2013; Morel et al., 2013], which receives preferential afferent projections from the basal ventromedial nucleus (VMB; also commonly referred to as the VPMpc), the dedicated gustatory and visceral relay nucleus of the thalamus [Pritchard et al., 1986]. The activation of this region of the insula is highlighted in neuroimaging studies of direct visceral stimulation, including cardiovascular arousal [Pollatos et al., 2007], heartbeat-evoked response [Park et al., 2014], gastric distension [Stephan et al., 2003; Wang et al., 2008], bladder filling [Mehnert et al., 2008], as well as vagus nerve stimulation [Kraus et al., 2007]. In this study, interoceptive attention, compared to exteroceptive attention, resulted in significantly greater activation of bilateral mid-to-posterior insula, in keeping with the results from prior human neuroimaging studies of interoception [Farb et al., 2013;

Pollatos et al., 2007; Simmons et al., 2013a]. Additionally, the subjects in this study also exhibited a significantly greater hemodynamic response to sweet vs. neutral tastants in bilateral dorsal mid-insula, specifically in the posterior short insular gyrus and overlying fronto-parietal operculum. The location of this gustatory responsive region supports prior evidence, utilizing multiple neuroimaging techniques, that this region is the location of primary gustatory cortex in humans [Kobayakawa et al., 1999; Nakamura et al., 2013; Ogawa et al., 2005; Small, 2010]. The dorsal mid-insula's putative role as primary gustatory and viscerosensory cortex is also supported by findings from meta-analyses of human neuroimaging studies [Kurth et al., 2010; Veldhuizen et al., 2011]. The coactivation of the dorsal mid-insula for tasks involving gustation and interoception observed within the present study provides experimental support solidifying the claim that these two sensory functions share their neural bases in the mid-insula.

The dorsal mid-insula's role seems to extend beyond one of primary sensory processing, as it supports the automatic retrieval of taste property inferences from visually presented food cues [Simmons et al., 2005; Van Der Laan et al., 2011]. The dorsal mid-insula is also involved in the hedonic response to food, as ratings of the perceived pleasantness of food items are directly related to activity within this region [Small, 2010], perhaps due to a direct connection to the ventral striatum [Fudge et al., 2005]. This relationship between insula activity and hedonic responses may be partially mediated by the effect of post-ingestive caloric signals to condition flavor preferences [De Araujo et al., 2013]. Additionally, recent neuroimaging evidence has demonstrated that the dorsal mid-insula's responsiveness to food cues is negatively correlated with peripheral levels of circulating glucose, a metabolic marker directly related to the body's available energy levels, suggesting that the response of gustatory cortex to food stimuli is dynamically modulated according to the current energy needs of the body [Simmons et al., 2013b]. This is consistent with a recent account of the role played by gustatory cortex in the integration of visual, olfactory, and physiological signals to regulate food intake [De Araujo et al., 2012]. The present finding that gustation and interoception jointly activate the dorsal mid-insula provides yet more support for De Araujo and colleagues' account.

Furthermore, previous neuroimaging studies have demonstrated that insula activity related to gustation is sensitive to nutritional factors such as dietary fat content and subjective feelings of hunger or satiety [Frank et al., 2012; Haase et al., 2009; Uher et al., 2006]. In particular, the response of the dorsal mid-insula to food pictures is also modulated by peripheral energy signals [Simmons et al., 2013b]. Therefore, it may be the case that differences in hunger or dietary nutrition could influence the differential representation of interoceptive and gustatory stimuli along the axis of the insula. For instance, when energy levels are low, peripheral hunger signals may result in increased

TABLE II. Cortical regions exhibiting significant activation in the interoceptive attention task

Side/Location	Peak Coordinates ^a				
<i>Interoception > Exteroception</i>	X	Y	Z	t_{19}	Volume (mm ³)
Right cuneus	+1	-73	+19	12.77	32032
Left superior frontal gyrus	-6	+11	+65	8.81	7395
Right middle temporal gyrus	+68	-28	+0	8.81	3728
Left superior temporal sulcus	-37	-53	+18	7.81	3630
Left superior parietal lobule	-12	-66	+53	7.16	2463
Left superior temporal gyrus	-59	+2	-3	7.16	2255
Right central sulcus	+23	-35	+54	8.26	2209
Left lateral orbitofrontal cortex	-27	+34	-5	8.61	1862
Right dorsal posterior insula	+34	-14	+20	6.61	1487
Left middle frontal gyrus	-33	+14	+56	7.21	1064
Right ventral posterior insula	+34	-21	+3	8.46	839
Left inferior frontal gyrus	-53	+25	+21	5.81	823
Left parahippocampal gyrus	-29	-37	-6	7.71	782
Left precentral gyrus	-20	-25	+53	8.36	601
Right lateral orbitofrontal cortex	+28	+31	-8	7.51	570
Left anterior temporal pole	-44	+14	-16	7.46	495
Right fusiform gyrus	+28	-52	-10	4.91	431
Right precentral gyrus	+25	-20	+63	5.31	319
Left ventral posterior insula	-33	-22	+3	6.61	281
Right precuneus	+0	-49	+49	5.26	262
Left dorsal mid-insula	-30	+1	+16	7.61	249
Left medial prefrontal cortex	-7	+58	+1	5.81	224
Left middle frontal gyrus	-39	+38	-4	5.36	215
Left precentral gyrus	-23	-10	+68	6.11	140
Right dorsal mid-insula ^b	+37	+1	+17	4.11	111
Left mid cingulate cortex	-13	-11	+44	4.46	94
<i>Exteroception > Interoception</i>					
Right precentral gyrus	+34	-3	+39	-8.66	12105
Right precuneus	+26	-54	+38	-9.95	11196
Right inferior occipital gyrus	+33	-75	-4	-10.26	10192
Left lingual gyrus	-30	-72	-7	-9.50	9844
Left medial frontal gyrus	+0	-3	+62	-11.27	6970
Right dorsal anterior insula	+35	+21	+1	-8.62	2536
Left precentral gyrus	-51	-7	+45	-7.25	1997
Left precuneus	-24	-60	+34	-6.63	807
Right posterior cingulate cortex	+3	-32	+43	-5.70	528
Right mid cingulate cortex	+4	-1	+26	-6.54	375
Left middle temporal gyrus	-28	-67	+28	-6.10	302
Left inferior parietal lobule	-40	-40	+47	-4.99	263

^aOverlapping activation for interoceptive and gustatory tasks observed within this cluster (see Fig. 2).

^bAll coordinates reported according to Talairach stereotaxic atlas.

response to gustatory stimuli, as well as the recruitment of other regions of the insula associated with the emotional and motivational aspects of feeding behavior. Findings to this effect would strongly support the idea that the gustatory-interoceptive overlap reported in the present study serves to help regulate energy intake and homeostasis. Future studies incorporating this experimental paradigm, along with direct measurements of metabolic indices and regulation of dietary food intake, will be needed to determine if this is the case.

Meta-analytic evidence of the functional organization of the insula has identified distinct functional zones within

the insula that support different categories of cognitive tasks [Kurth et al., 2010]. Dorsal anterior insula is considered to be a “cognitive-attentional” region, ventral anterior insula a ‘social-emotional’ region, and mid-to-posterior insula are involved in ‘sensori-motor’ and ‘chemical sensory’ processing [Kurth et al., 2010]. This functional topography is supported by experimental evidence which identified that specific regions of the insula are selectively activated by stimuli with these qualitatively different characteristics [Simmons et al., 2013a]. These findings suggest the possibility that differences in the cognitive demands of the interoceptive and exteroceptive conditions of the IA

TABLE III. Cortical regions exhibiting significant activation in the gustatory mapping task

Side/Location	Peak Coordinates ^a			t_{19}	Volume (mm ³)
	X	Y	Z		
<i>Sweet > Neutral Tastant</i>					
L Postcentral Gyrus	-58	-18	+32	9.57	1504
R Postcentral Gyrus	+62	-28	+25	6.85	1307
L Precentral Gyrus	-58	-2	+25	6.46	1263
R Precentral Gyrus	+56	+0	+38	6.21	881
R Dorsal Mid-Insula ^b	+37	-6	+17	6.50	397
L Dorsal Mid-Insula	-34	-9	+15	6.12	366
L Inferior Frontal Gyrus	-47	-3	+15	5.71	114

^aOverlapping activation for interoceptive and gustatory tasks observed within this cluster (see Figure 2).

^bAll coordinates reported according to Talairach stereotaxic atlas.

task (i.e., intensity evaluation vs. counting) may influence the degree and extent of their functional activation along the axis of the insula. Future designs of the IA paradigm will need to take into account these differences in task demands by, for example, incorporating intensity evaluation into the exteroception condition.

CONCLUSION

The findings presented here are novel and important precisely because they demonstrate, for the first time in humans, the overlapping cortical representation of gustatory and interoceptive signals. This common neural representation in the insula shared by these two sensory modalities suggests a potential role for the mid-insula in the integration of physiological and taste-related information. While the results from the present study support the notion of multimodal sensory processing within the dorsal mid-insula, they are unable to directly demonstrate whether the overlapping activations within this region reflect the responses of distinct or shared populations of neurons. Although strong evidence for the latter comes from translational evidence in rodents demonstrating the activation of individual insular neurons by both gustatory and interoceptive stimuli [Hanamori et al., 1998], further research is needed to determine if this is also the case within the human dorsal mid-insula. Though current limits on the effective image resolution of human neuroimaging methods may hinder the ability to directly examine the response properties of individual neurons, more indirect techniques such as fMRI-Adaptation [Grill-Spector and Malach, 2001] might be employed to discriminate between the responses of different neuronal populations at the subvoxel level.

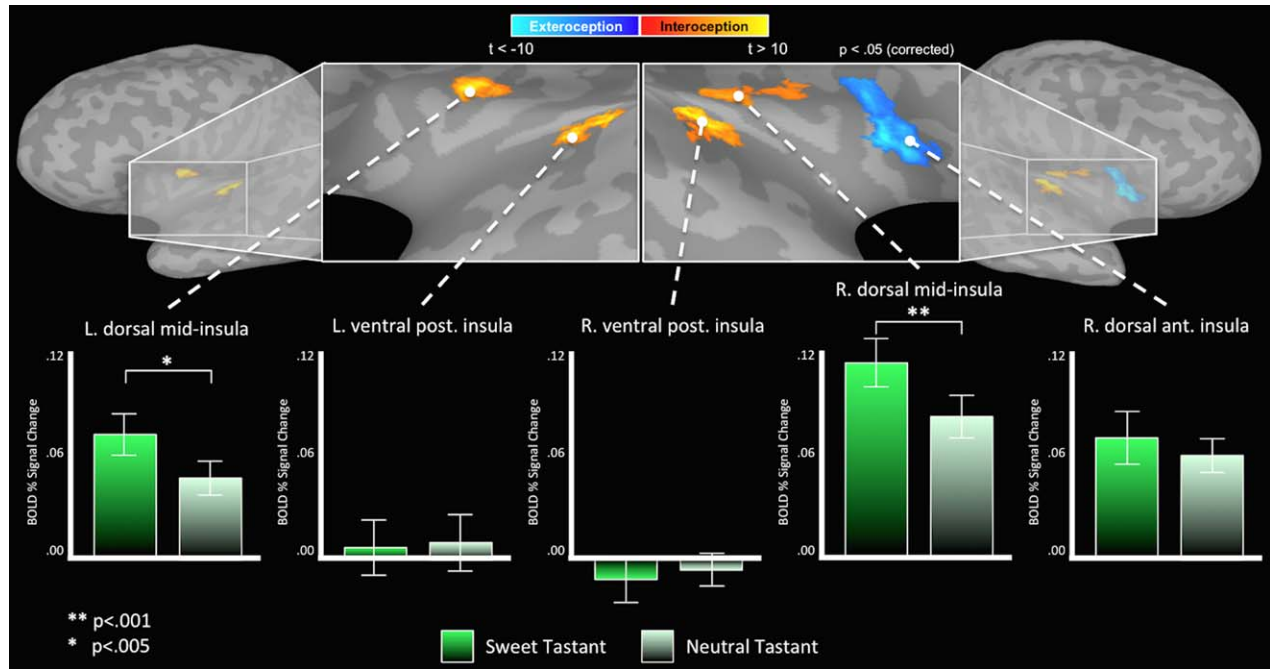


Figure 3.

Gustatory ROI analyses. The regions of the insula identified in Figure 2a were used to examine the gustatory response profile across posterior, mid, and anterior regions of the insula. Only the bilateral dorsal mid-insula exhibited both a significant response to sweet and neutral tastants individually, as well as a significant difference between those responses.

TABLE IV. ROI analyses

Insula region	Sweet tastant			Neutral tastant			Sweet-neutral ^a	
	% signal change			% signal change			% signal change	
<i>Interoception > Exteroception</i>	<i>Mean(SD)</i>	<i>t(19)</i>	<i>P</i>	<i>Mean(SD)</i>	<i>t(19)</i>	<i>P</i>	<i>t(19)</i>	<i>P</i>
L Dorsal Mid-Insula	0.07(0.05)	5.91	<0.001*	0.05(0.04)	4.85	<0.001*	3.17	<0.005*
L Ventral Posterior Insula	0.01(.07)	0.37	0.71	0.01(0.07)	0.59	0.41	−0.63	0.54
R Dorsal Mid-Insula	0.11(.06)	8.28	<0.001*	0.08(0.05)	6.65	<0.001*	4.34	<0.001*
R Ventral Posterior Insula	−0.01(.06)	−0.91	0.37	−0.01(0.04)	−0.75	0.46	−0.51	0.62
<i>Exteroception > Interoception</i>								
R Dorsal Anterior Insula	0.07(.07)	4.58	<0.001*	0.06(0.04)	6.01	<.001*	1.26	0.22

^a*t*-values were computed using paired *t*-tests.

*Statistically significant after FDR-correction for multiple comparisons at $p_{\text{corrected}} < 0.05$ [Benjamini and Hochberg, 1995].

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CONFLICT OF INTEREST

The authors declare no competing financial interests.

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